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Code No: 138GW

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
RENEWABLE ENERGY SOURCES
(Common to CE, EEE, ME, ECE, CSE)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Distinguish between renewable energy and non-renewable energy resources.
b) What are the advantages and limitations of renewable energy resources? [8+7]

2.a) Explain the CO_2 emission reduction potential assessment using renewable energy in India.
b) Discuss the importance of renewable energy in the fight against global climate change. [7+8]

3.a) Define: (i) Latitude angle (ii) Altitude angle (iii) Hour angle (iv) Azimuth angle (v) Collector slope angle.
b) What is the difference between a pyrheliometer and pyranometers? [7+8]

4.a) List out the advantages and disadvantages of solar PV systems.
b) With a neat sketch, explain any one type of thermal energy storage system. [8+7]

5.a) What is the difference between wind monitoring stations and wind farms?
b) State and briefly explain the factors that determine the output power from wind energy. [8+7]

6.a) List out the wind energy potential locations in India and give the details of installed capacity of wind energy.
b) Discuss the importance of wind resource assessment. [8+7]

7.a) Compare and contrast biomass and biogas.
b) Briefly discuss the factors affecting anerobic digestion. [8+7]

8. Write a short note on:
a) Small hydro resources
b) Geothermal energy
c) Tidal energy
d) Wave energy. [15]

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